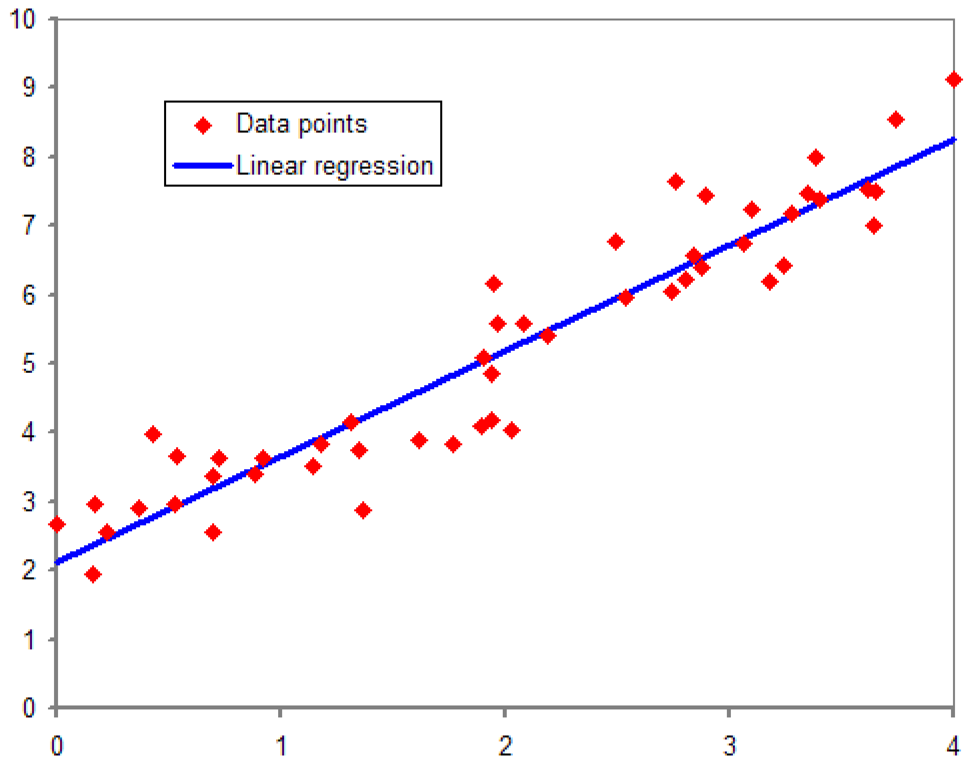
Regression

What is regression?

- Regression is a supervised learning technique used to model the relationship between input variables (features) and a continuous output variable.
- In regression problems there will be:
  - A dependent variable (called the 'outcome' or 'response' variable) and
  - one or more independent variables (called 'predictors', 'covariates', 'explanatory variables' or 'features').



Types of Regression

- ***Linear Regression***
  - ***Simple Linear Regression*** or ***Univariate Linear Regression***.
    - Only one independent variable
  - ***Multiple Linear Regression*** or ***Multivariate Linear Regression***.
    - More than one independent variable
- ***Non-linear regression***
    - commonly used for more complicated data sets in which the dependent and independent variables show a nonlinear relationship

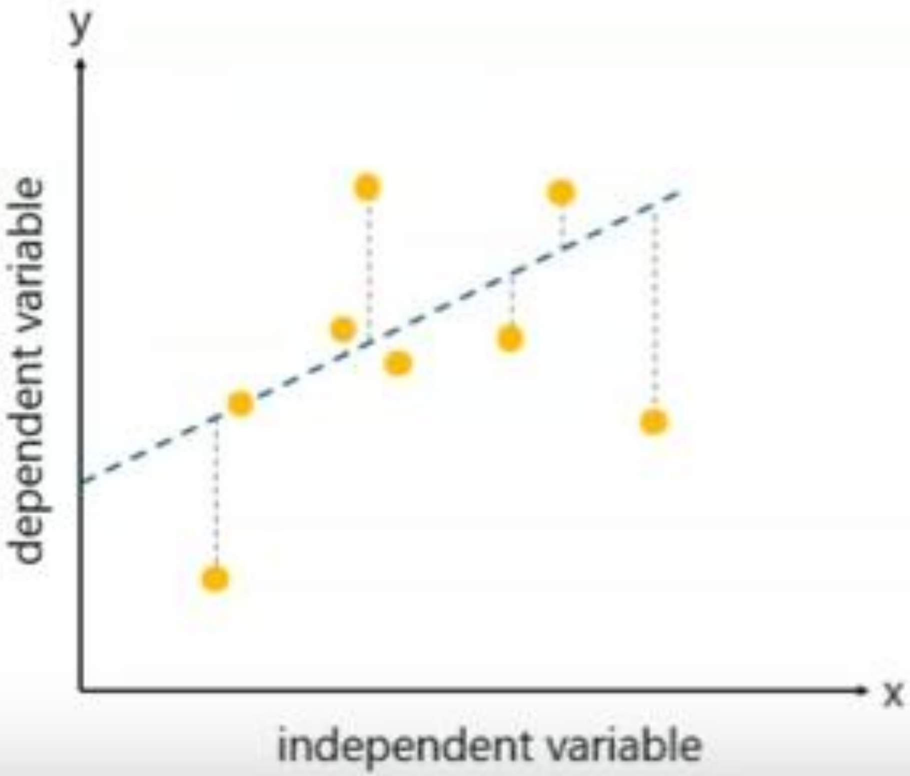
Linear Regression

Dependent and Independent variable

Expression	Independent	Dependent
$y = 3 + 2x$	x	y
$y = x^2 - 2x$	x	y
$z = 5x^2 + 8y^3$	x, y	z

Linear Regression

Linear Regression is a method to predict dependent variable (Y) based on values of independent variables (X). It can be used for the cases where we want to predict some continuous quantity.



- *Dependent variable (Y):*

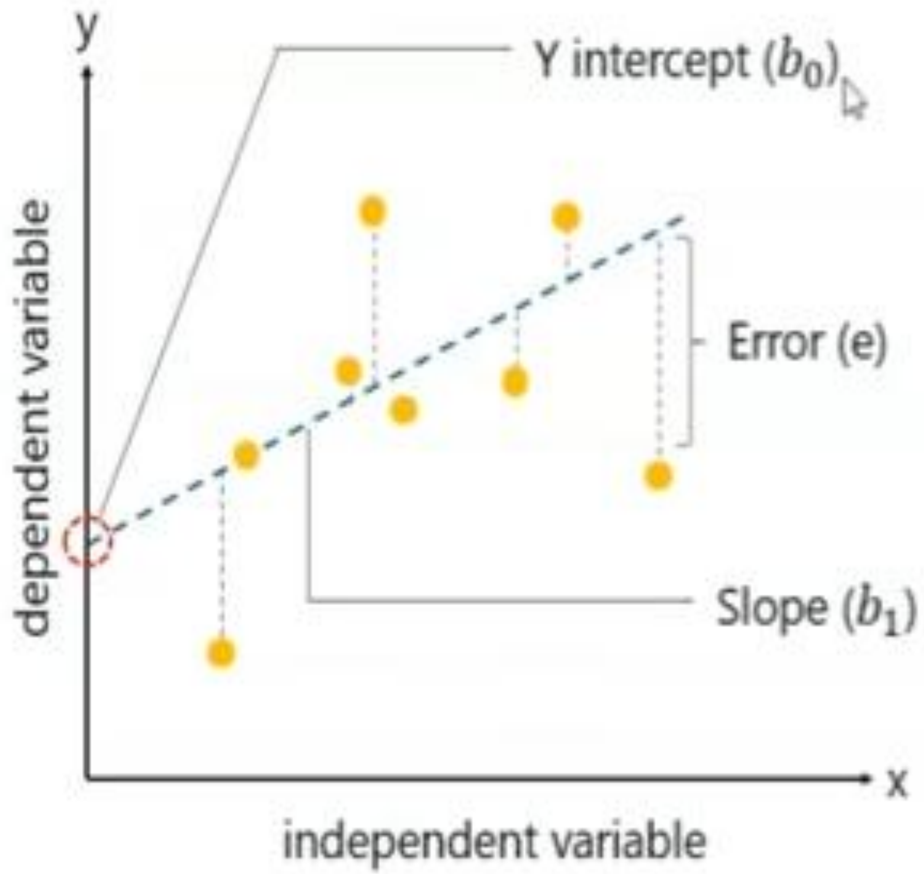
The response variable who's value needs to be predicted.

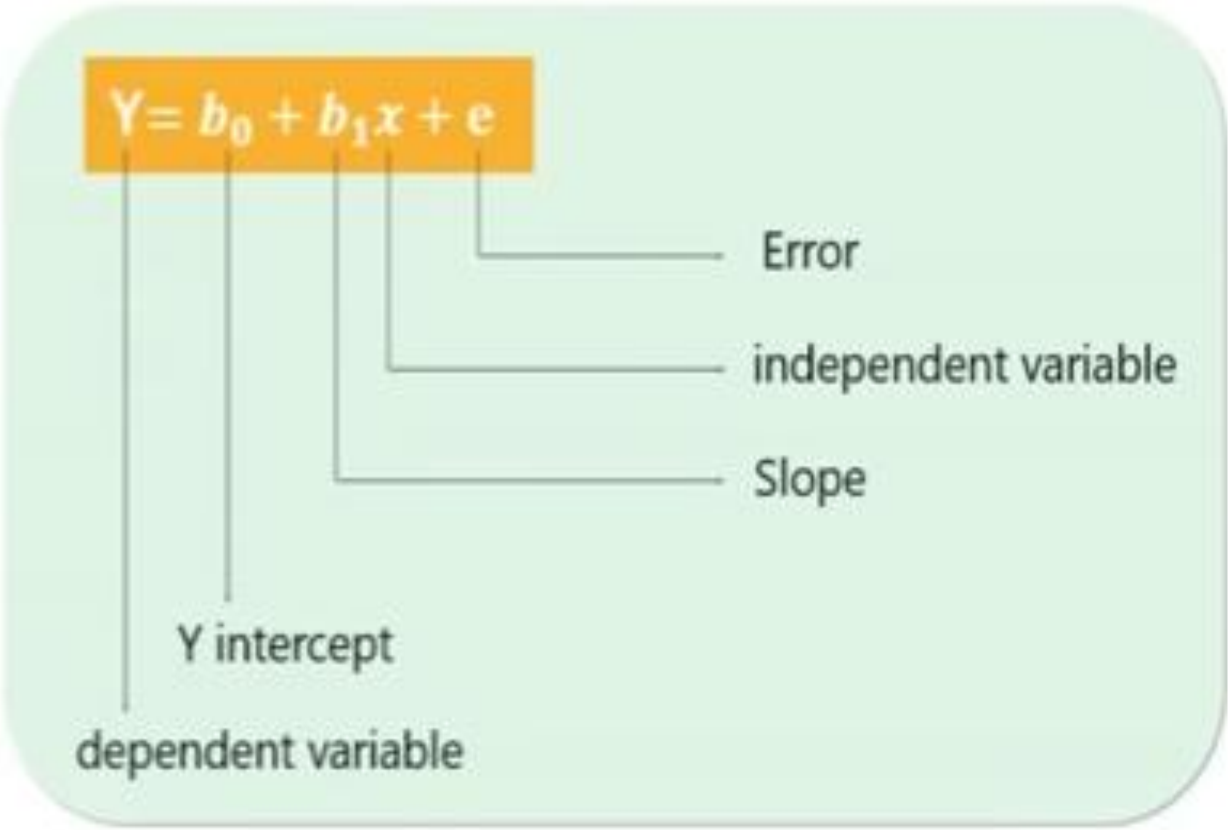
- *Independent variable (X):*

The predictor variable used to predict the response variable.

The following equation is used to represent a linear regression model:

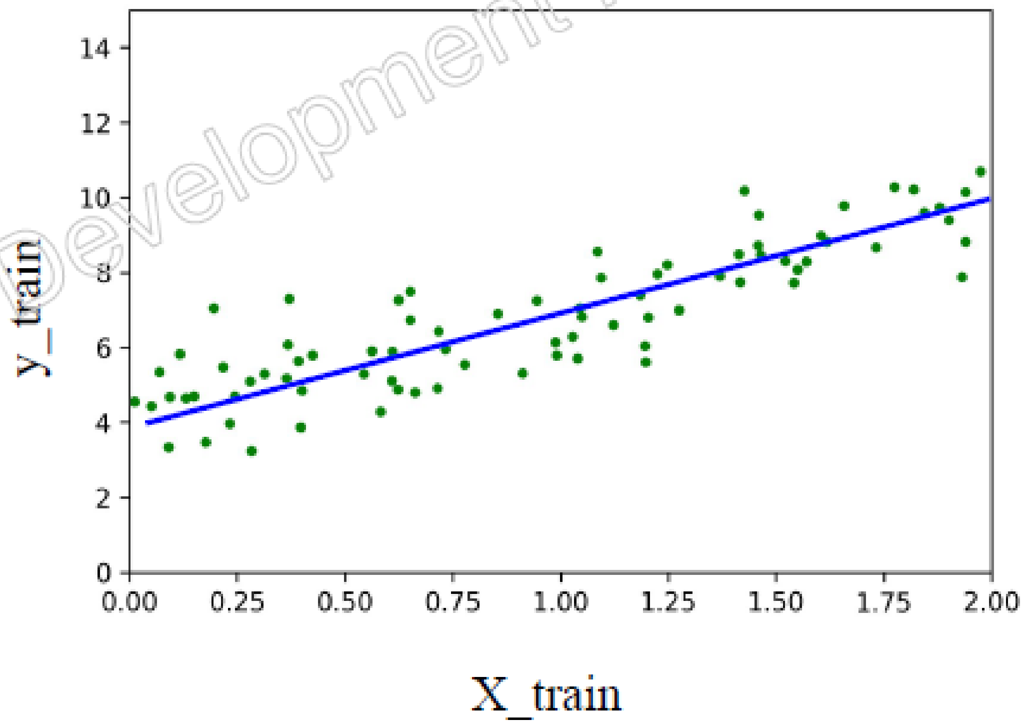
$$Y = b_0 + b_1 x + e$$


y
Y intercept (b_0)
Error (e)
Slope (b_1)
dependent variable
x
independent variable


$Y = b_0 + b_1x + e$
Error
independent variable
Slope
Y intercept
dependent variable

Simple Linear Regression

We can obtain here a **straight line** which passes close to as many points as possible.



What parameters are required to represent a straight line?

y-intercept
slope

Equation of a straight line:

$$y = c + mx$$

Example

X (independent variable)	Y (Dependent variable)
1	2
2	3
3	5
4	4
5	6

We fit a line: $y = c + mx$

$$m = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$$

$$c = \frac{\sum y - m\sum x}{n}$$

x	y	x^2	xy
1	2	1	2
2	3	4	6
3	5	9	15
4	4	16	16
5	6	25	30

$$\sum x = 15, \quad \sum y = 20$$

$$\sum x^2 = 55, \quad \sum xy = 69$$

$$n = 5$$

$$m = \frac{5(69) - 15(20)}{5(55) - 15^2}$$

$$m = \frac{345 - 300}{275 - 225}$$

$$m = \frac{45}{50} = 0.9$$

$$c = \frac{20 - 0.9(15)}{5}$$

$$c = \frac{20 - 13.5}{5}$$

$$c = \frac{6.5}{5} = 1.3$$

Final Regression Equation

$$\boxed{y = 1.3 + 0.9x}$$

Predicted values

For $x = 3$:

$y = 1.3 + 0.9(3) = 4.0$

- Number of man-hours and the corresponding productivity (in units) are furnished below. Fit a simple linear regression equation using method of least squares. What will be the productivity if man-hours = 12.

Man-hours	3.6	4.8	7.2	6.9	10.7	6.1	7.9	9.5	5.4
Productivity (in units)	9.3	10.2	11.5	12	18.6	13.2	10.8	22.7	12.7

Man-hours x_i	Productivity y_i	x_i^2	$x_i y_i$
3.6	9.3	12.96	33.48
4.8	10.2	23.04	48.96
7.2	11.5	51.84	82.8
6.9	12	47.61	82.8
10.7	18.6	114.49	199.02
6.1	13.2	37.21	80.52
7.9	10.8	62.41	85.32
9.5	22.7	90.25	215.65
5.4	12.7	29.16	66.42
$\sum_{i=1}^{9} x_i = 62.1$	$\sum_{i=1}^{9} y_i = 121$	$\sum_{i=1}^{9} x_i^2 = 468.97$	$\sum_{i=1}^{9} x_i y_i = 894.97$

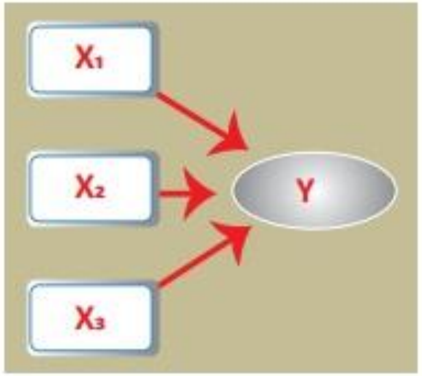
$\hat{Y} = 2.88 + 1.53x$

- A high school statistics teacher wants to know if the number of hours a student spends studying can predict their final exam score. The teacher collects data from a small sample of four students, recording the hours studied (x) and the exam scores (y). Using method of least squares, Predict the exam score for a student who studies for **5 hours**.

Student	Hours Studied (x)	Exam Score (y)
1	1	42
2	2	50
3	3	57
4	4	65
5	6	78
6	7	85
7	8	92

Multiple Linear Regression

Multiple linear regression uses two or more independent variables to estimate the value(s) of the response variable (*Y*).



The general form of the multiple linear regression equation is

$Y = b_o + b_1X_1 + b_2X_2 + b_3X_3 + \ldots + b_tX_t + e$

b_o = y-intercept {a constant value}

b_1 = slope of Y with variable x_1 holding the variables x_2, x_3, ..., x_t effects constant

b_t = slope of Y with variable x_t holding all other variables' effects constant

Numerical Example

y	X_1	X_2
140	60	22
155	62	25
159	67	24
179	70	20
192	71	15
200	72	14
212	75	14
215	78	11

$$Y = b_o + b_1X_1 + b_2X_2$$

$$b_0 = \bar{y} - b_1\bar{X}_1 - b_2\bar{X}_2$$

$$b_1 = [(\Sigma x_2^2)(\Sigma x_1 y) - (\Sigma x_1 x_2)(\Sigma x_2 y)] \; / \; [(\Sigma x_1^2)(\Sigma x_2^2) - (\Sigma x_1 x_2)^2]$$

$$b_2 = [(\Sigma x_1^2)(\Sigma x_2 y) - (\Sigma x_1 x_2)(\Sigma x_1 y)] \; / \; [(\Sigma x_1^2)(\Sigma x_2^2) - (\Sigma x_1 x_2)^2]$$

Procedure

Step 1: Calculate X_1^2, X_2^2, X_1y, X_2y and X_1X_2.

Step 2: Calculate Regression Sums.

$$\Sigma x_1^2 = \Sigma X_1^2 - (\Sigma X_1)^2 / n$$
$$\Sigma x_2^2 = \Sigma X_2^2 - (\Sigma X_2)^2 / n$$
$$\Sigma x_1 y = \Sigma X_1 y - (\Sigma X_1 \Sigma y) / n$$
$$\Sigma x_2 y = \Sigma X_2 y - (\Sigma X_2 \Sigma y) / n$$
$$\Sigma x_1 x_2 = \Sigma X_1 X_2 - (\Sigma X_1 \Sigma X_2) / n$$

Step 3: Calculate b_0, b_1, and b_2.

y	X_1	X_2
140	60	22
155	62	25
159	67	24
179	70	20
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200	72	14
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	y	X_1	X_2
	140	60	22
	155	62	25
	159	67	24
	179	70	20
	192	71	15
	200	72	14
	212	75	14
	215	78	11
Mean	181.5	69.375	18.125
Sum	1452	555	145

	X_1^2	X_2^2	X_1y	X_2y	X_1X_2
	3600	484	8400	3080	1320
	3844	625	9610	3875	1550
	4489	576	10653	3816	1608
	4900	400	12530	3580	1400
	5041	225	13632	2880	1065
	5184	196	14400	2800	1008
	5625	196	15900	2968	1050
	6084	121	16770	2365	858
Sum	38767	2823	101895	25364	9859
Reg Sums	263.875	194.875	1162.5	-953.5	-200.375

$Y = b_o + b_1X_1 + b_2X_2$

$b_0 = \overline{y} - b_1\overline{X}_1 - b_2\overline{X}_2$

$b_1 = [(\Sigma x_2^2)(\Sigma x_1y) - (\Sigma x_1x_2)(\Sigma x_2y)] \;/\; [(\Sigma x_1^2)(\Sigma x_2^2) - (\Sigma x_1x_2)^2]$

$b_2 = [(\Sigma x_1^2)(\Sigma x_2y) - (\Sigma x_1x_2)(\Sigma x_1y)] \;/\; [(\Sigma x_1^2)(\Sigma x_2^2) - (\Sigma x_1x_2)^2]$

$\Sigma x_1^2 = \Sigma X_1^2 - (\Sigma X_1)^2 / n = 38{,}767 - (555)^2 / 8 = \mathbf{263.875}$
$\Sigma x_2^2 = \Sigma X_2^2 - (\Sigma X_2)^2 / n = 2{,}823 - (145)^2 / 8 = \mathbf{194.875}$
$\Sigma x_1y = \Sigma X_1y - (\Sigma X_1\Sigma y) / n = 101{,}895 - (555*1{,}452) / 8 = \mathbf{1{,}162.5}$
$\Sigma x_2y = \Sigma X_2y - (\Sigma X_2\Sigma y) / n = 25{,}364 - (145*1{,}452) / 8 = \mathbf{-953.5}$
$\Sigma x_1x_2 = \Sigma X_1X_2 - (\Sigma X_1\Sigma X_2) / n = 9{,}859 - (555*145) / 8 = \mathbf{-200.375}$

$\mathbf{b_1}$ = [(194.875)(1162.5) – (-200.375)(-953.5)] / [(263.875) (194.875) – $(-200.375)^2$] = **3.148**

$\mathbf{b_2}$ = [(263.875)(-953.5) – (-200.375)(1152.5)] / [(263.875) (194.875) – $(-200.375)^2$] = **-1.656**

$\mathbf{b_0}$ = 181.5 – 3.148(69.375) – (-1.656)(18.125) = **-6.867**

$\hat{y} = b_0 + b_1*x_1 + b_2*x_2$

$\boldsymbol{\hat{y} = -6.867 + 3.148x_1 – 1.656x_2}$

How to Interpret a Multiple Linear Regression Equation

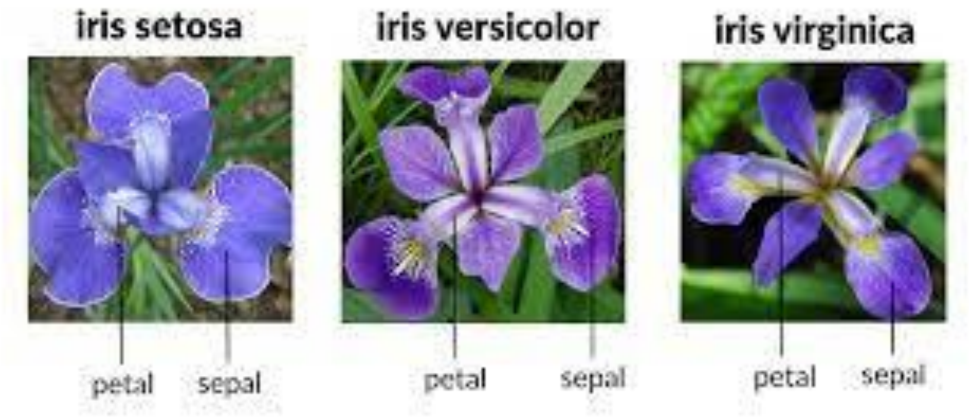
$\mathbf{b_0 = -6.867}$. When both predictor variables are equal to zero, the mean value for y is -6.867.

$\mathbf{b_1 = 3.148}$. A one unit increase in x_1 is associated with a 3.148 unit increase in y, on average, assuming x_2 is held constant.

$\mathbf{b_2 = -1.656}$. A one unit increase in x_2 is associated with a 1.656 unit decrease in y, on average, assuming x_1 is held constant.

How machine learns?

- Training set, Validation Set, Test set
- Features and feature selection



Training Set for classification

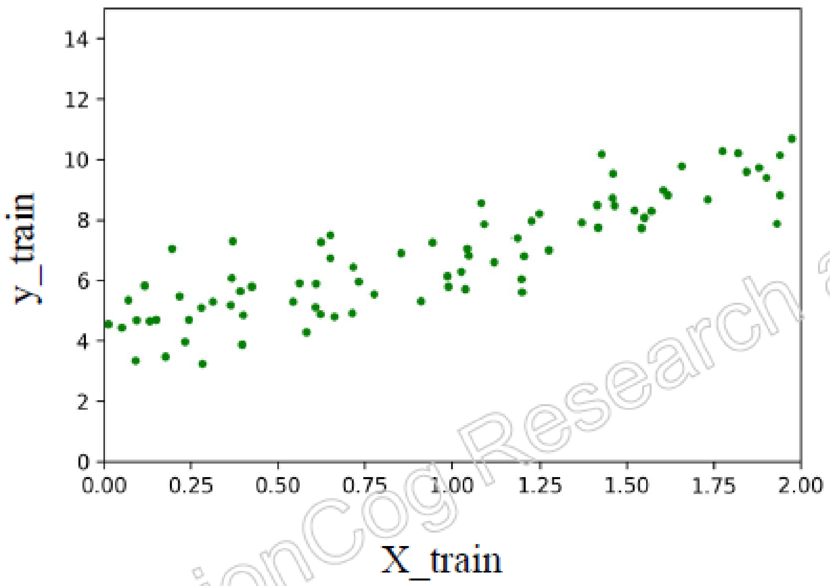
Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
1	5.1	3.5	1.4	0.2	Iris-setosa
2	4.9	3	1.4	0.2	Iris-setosa
3	4.7	3.2	1.3	0.2	Iris-setosa
4	4.6	3.1	1.5	0.2	Iris-setosa
5	5	3.6	1.4	0.2	Iris-setosa
6	5.4	3.9	1.7	0.4	Iris-setosa
7	4.6	3.4	1.4	0.3	Iris-setosa
8	5	3.4	1.5	0.2	Iris-setosa
9	4.4	2.9	1.4	0.2	Iris-setosa
10	4.9	3.1	1.5	0.1	Iris-setosa
51	7	3.2	4.7	1.4	Iris-versicolor
52	6.4	3.2	4.5	1.5	Iris-versicolor
53	6.9	3.1	4.9	1.5	Iris-versicolor
54	5.5	2.3	4	1.3	Iris-versicolor
55	6.5	2.8	4.6	1.5	Iris-versicolor
56	5.7	2.8	4.5	1.3	Iris-versicolor
57	6.3	3.3	4.7	1.6	Iris-versicolor
58	4.9	2.4	3.3	1	Iris-versicolor
59	6.6	2.9	4.6	1.3	Iris-versicolor
60	5.2	2.7	3.9	1.4	Iris-versicolor
102	5.8	2.7	5.1	1.9	Iris-virginica
103	7.1	3	5.9	2.1	Iris-virginica
104	6.3	2.9	5.6	1.8	Iris-virginica

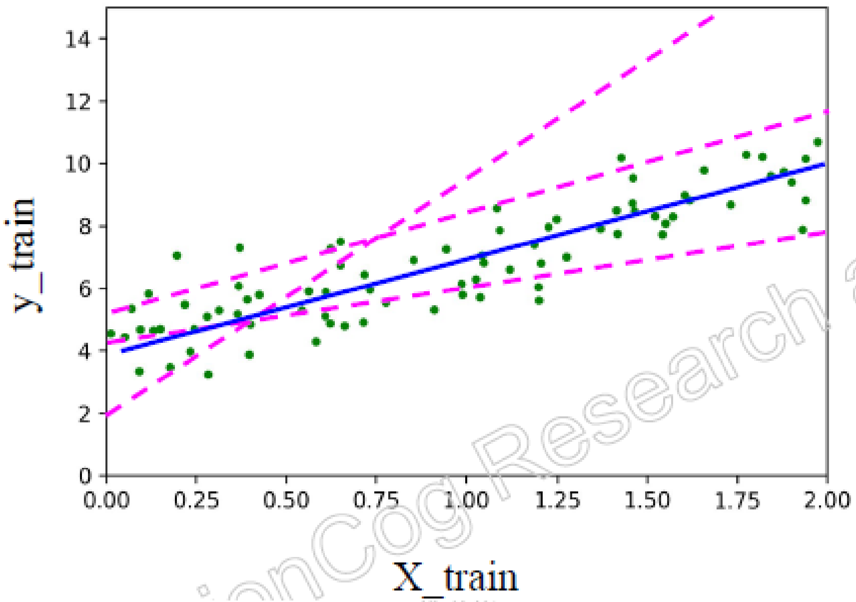
How machine learns?

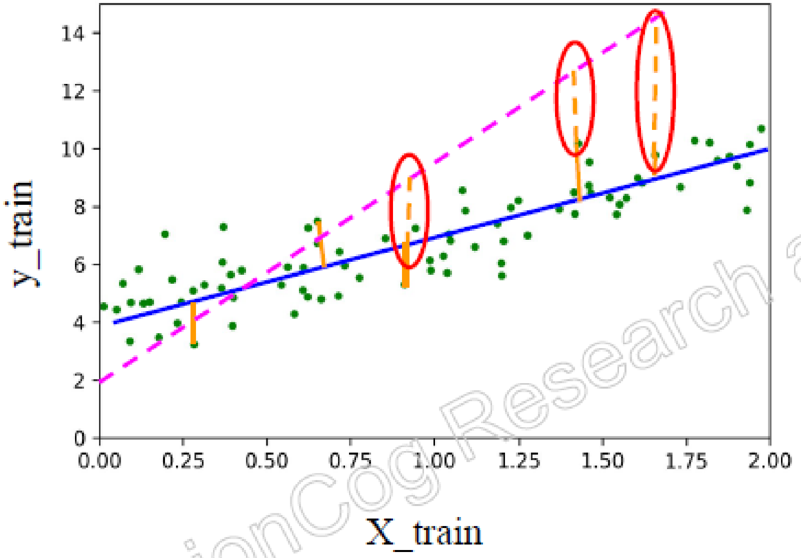
Training set, Validation Set, Test set

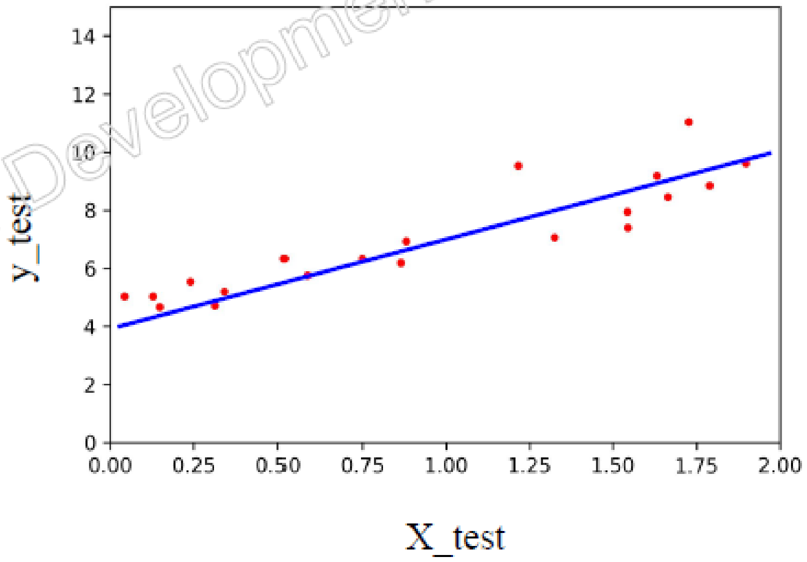
X1: Area (sq ft)	X2: Bedrooms	X3: Age (years)	X4: Distance to City (km)	Y: Price (Lakhs)
800	2	20	15	45
950	2	15	12	55
1100	3	10	10	70
1200	3	8	8	78
1350	3	5	6	90
1500	4	3	5	105
1650	4	2	4	120
1800	4	1	3	135
2000	5	1	2	155
2200	5	0	1	180

How machine learns?



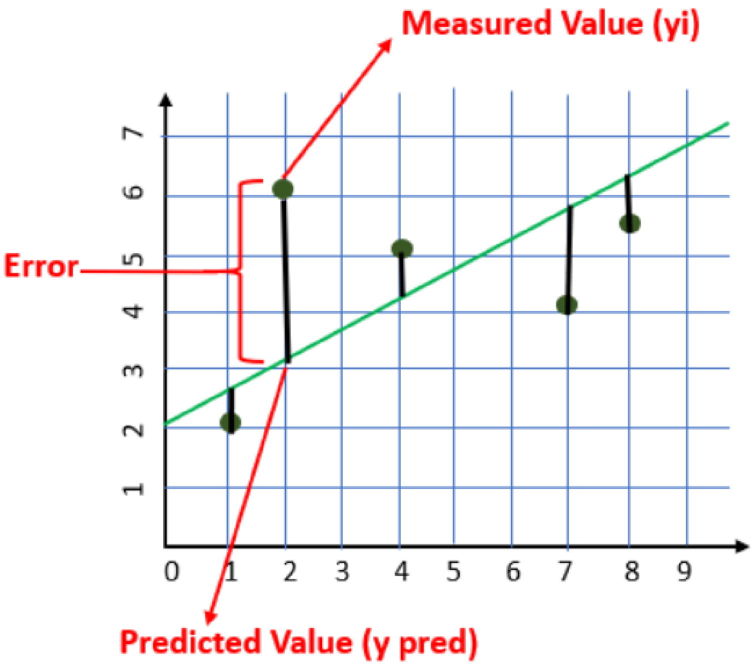






Cost/Loss Function: Estimating the error in prediction

- Machines learn by considering loss function.
- Cost functions (or loss functions) in machine learning measure the error between the model's predictions and the actual target values.
- If predictions deviates too much from actual results, loss function will be a very large number.
- Gradually, with the help of some optimization function, loss function learns to reduce the error in prediction.
- The choice of cost function depends on the task (classification, regression, etc.).



Loss functions for regression

1. Mean Bias Error (MBE)

$$MBE = \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)}{n}$$

- ➢Mean Bias Error takes the actual difference between the target and the predicted value, and not the absolute difference.
- ➢One has to be cautious as the positive and the negative errors could cancel each other out, which is why it is one of the lesser-used loss functions

2. Mean Absolute Error (MAE)

$$MAE = \frac{\sum_{i=1}^{n} | \; y_i - \hat{y}_i \; |}{n}$$

- ➢MAE takes the average sum of the absolute differences between the actual and the predicted values.
- ➢Regression problems may have variables that are not strictly Gaussian in nature due to the presence of outliers (values that are very different from the rest of the data).

➢Mean Absolute Error would be an ideal option in such cases because it does not take into account the direction of the outliers

3. Sum of Squares Error (SSE)

$$SSE = \Sigma(\hat{y}_i - y_i)^2$$

The sum of squared differences between predicted data points (ŷi) and observed data points (yi).

4 Mean Squared Error (MSE)

- Mean Squared Error is the average of the squared differences between the actual and the predicted values.

$$\mathrm{MSE} = \frac{1}{n} \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$$

- Mean Squared Error (also called L2 loss) is almost every data scientist’s preference when it comes to loss functions for regression.
- This is because most variables can be modeled into a Gaussian distribution.
- In MSE, since each error is squared, it helps to penalize even small deviations in prediction when compared to MAE.
- But if our dataset has outliers that contribute to larger prediction errors, then squaring this error further will magnify the error many times more and also lead to higher MSE error.
- Hence we can say that it is less robust to outliers

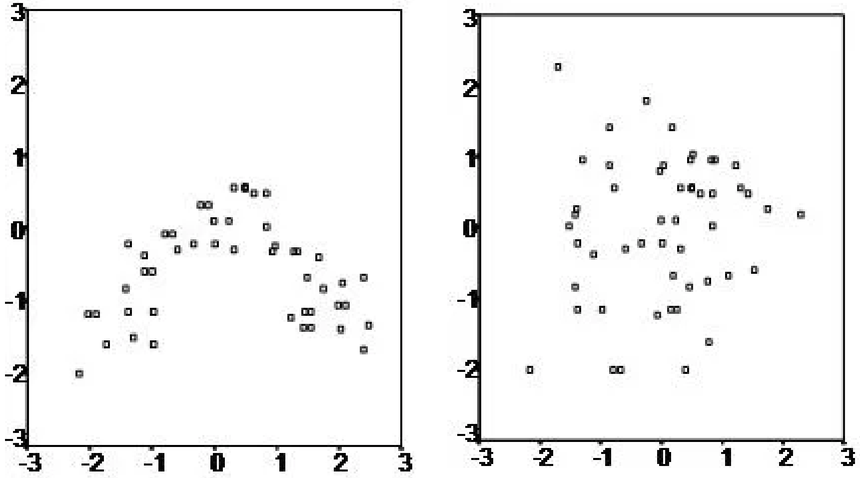
Assumptions of Linear Regression

- The regression has five key assumptions:
  - Linear relationship
  - Independence
  - Multivariate normality
  - No or little multicollinearity
  - No auto-correlation
  - Homoscedasticity (or no Heteroscedasticity)

  In Linear_regression the sample size rule of thumb is that the regression analysis requires at least 20 cases per independent variable in the analysis.

Continued…

- Linear Relationship
  - **linear regression needs the relationship between the independent and dependent variables to be linear.**
  - **The linearity assumption can best be tested with scatter plots, the following two examples depict two cases, where no and little linearity is present.**
  - **It is also important to check for outliers since linear regression is sensitive to outlier effects.**



no linearity

- **Independence:** The observations are independent of each other.
- **Normality:** The residues/errors follow a normal distribution.
- **No multicollinearity:** The independent variables are not highly correlated with each other.
  - When the independent variables are highly correlated to each other then the variables are said to be multicollinear.
  - Many types of regression techniques assumes multicollinearity should not be present in the dataset. It is because it causes problems in ranking variables based on its importance. Or it makes job difficult in selecting the most important independent variable (factor/ Feature).

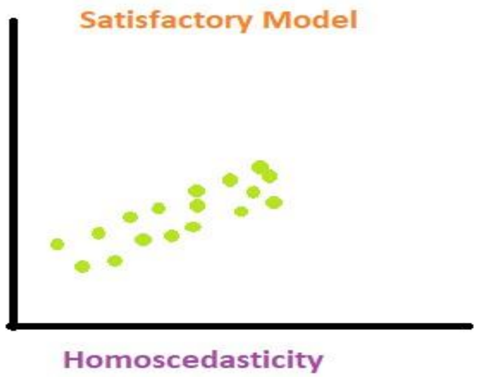
- **No autocorrelation:**
  - Autocorrelation refers to the degree of correlation of the same variables between two successive time intervals. It measures how the lagged version of the value of a variable is related to the original version of it in a time series
  - Autocorrelation occurs when the residuals/ errors are not independent from each other.
    - For instance, this typically occurs in stock prices, where the price is not independent from the previous price.

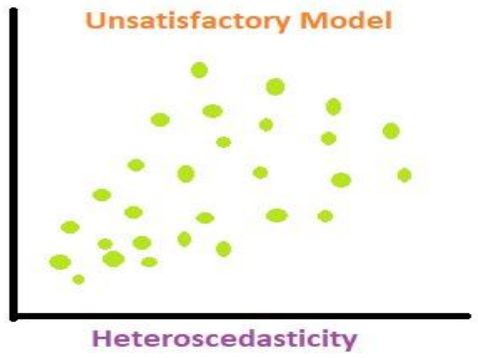
No Heteroscedasticity

- **Homoscedasticity:** The variance of the errors is constant across all levels of the independent variables.

- When dependent variable's variability is not equal across values of an independent variable, it is called heteroscedasticity.

- Example -As one's income increases, the variability of food consumption will increase. A poorer person will spend a rather constant amount by always eating inexpensive food; a wealthier person may occasionally buy inexpensive food and at other times eat expensive meals. Those with higher incomes display a greater variability of food consumption.





- **Heteroscedasticity** refers to condition in which the variance of error term or the residual term in a regression model varies. As you can see in the diagram, in case of homoscedasticity, the data points are equally scattered while in case of heteroscedasticity the data points are not equally scattered.